

Héctor Masip Ardevol

Research Engineer
ZisK

CONTACT

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LinkedIn

SKILLS

Programming

Rust
Python / Sage
Lean 4
JavaScript / TS

ZK & Blockchain

PIL / PIL2
Circom
zkASM
Solidity

LANGUAGES

Spanish Native
Catalan Native
English Fluent

INTERESTS

Math
Probabilistic Proofs
Formal Methods
Distributed Systems

I build systems that prove computation without revealing secrets. Research engineer working on zero-knowledge proof systems, from the math to the prover code: currently building ZisK, a high-performance zkVM; previously the Polygon zkEVM proving stack.

EXPERIENCE

Jun 2025 – Present
Research Engineer, ZisK
Building a high-performance zkVM.

May 2023 – Jun 2025
Research Engineer, Polygon Labs
Polygon zkEVM: arithmetization with the extended AIR (eAIR), and the integrity of the proving system, a composition of an inner eSTARK with an outer fflonK.

2021 – 2023
Doctoral Researcher, Polytechnic University of Catalonia
Research on zero-knowledge proofs for blockchain scalability, with a focus on zkEVMs, in the Department of Network Engineering. Funded by the FI-SDUR 2021 grant (Generalitat de Catalunya).

2021
Research Staff Member, Polytechnic University of Catalonia
Research member of the ARPASAT project.

2020
Research Collaborator, Polytechnic University of Catalonia & Codegram
Analysis, study and implementation of a post-quantum e-voting protocol.

EDUCATION

2021 – 2023
Doctoral Studies, Applied Cryptography, Polytechnic University of Catalonia
Department of Network Engineering. Left the program before completing the degree.

Summer 2022
Summer Residency, 0xPARC
Research residency on applied zero-knowledge cryptography in New York.

2019 – 2020
MSc, Advanced Mathematics and Mathematical Engineering, Polytechnic University of Catalonia
Thesis on linkable attribute-based signatures.

2015 – 2019
BSc, Mathematics, University of Alicante
Focus on discrete mathematics. Thesis on the subgraph isomorphism problem.

PUBLICATIONS

2026
Security of ZisK
Héctor Masip Ardevol. *Technical report, ZisK.*

2026
ZisK Whitepaper
ZisK Core Team. *Technical report, ZisK.*

2024
eSTARK: Extending STARKs with Arguments
Héctor Masip Ardevol, Marc Guzmán Albiol, Jordi Baylina Melé, Jose Luis Muñoz Tapia. *Designs, Codes and Cryptography 92(11).*

2022
RapidUp: Multi-Domain Permutation Protocol for Lookup Tables
Héctor Masip Ardevol, Jordi Baylina Melé, Daniel Lubarov, Jose Luis Muñoz Tapia. *IACR ePrint 2022/1050.*

2022
PlonKup: Reconciling PlonK with plookup
Luke Pearson, Joshua Fitzgerald, **Héctor Masip Ardevol**, Marta Bellés Muñoz, Jose Luis Muñoz Tapia. *IACR ePrint 2022/086.*

2020
Linkable Attribute-Based Signature
Héctor Masip Ardevol. *Master's thesis, Polytechnic University of Catalonia (UPC). Advisors: M. Paz Morillo Bosch and Ramiro Martínez Pinilla.*

TEACHING

2021, 2022
Lecturer, Cryptography, Master's in Blockchain Technologies, UPC School

AWARDS & GRANTS

2021 – 2023
FI-SDUR 2021 predoctoral grant
Generalitat de Catalunya. Funding for doctoral research at UPC.

2021
Honorable Mention, Premio Leonardo Torres Quevedo 2020 en Criptología y Seguridad de la Información
LTQ Awards. For the master's thesis on linkable attribute-based signatures.